ME315a_Lab03_LuizCelin

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Setup

QUESTÃO 1

```
airports <- read_csv(
  "airports.csv",
  col_types = cols_only(
    IATA_CODE = col_character(),
    CITY = col_character(),
    STATE = col_character(),
    LATITUDE = col_double(),
    LONGITUDE = col_double()
  )
)
```

```
airlines <- read_csv("airlines.csv")
```

```
getStats <- function(input, pos) {
  input %>%
    filter(!startsWith(DESTINATION_AIRPORT, "1")) %>%
    filter(!is.na(DESTINATION_AIRPORT) &
             !is.na(ARRIVAL_DELAY)) %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      n_voos = n()
    )
}
```

```
flights <- read_csv_chunked(
  "flights.csv.zip",
  callback = DataFrameCallback$new(getStats),
  chunk_size = 1e6,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)
```

```
flights <- flights %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    total_atrasos = sum(soma_atrasos),
    total_voos = sum(n_voos)
  ) %>%
  mutate(
    atraso_medio = total_atrasos / total_voos
  )
```

QUESTÃO 2

```
flights %>%
  anti_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) %>%
  select(DESTINATION_AIRPORT)
```

```
# A tibble: 0 x 1
# i 1 variable: DESTINATION_AIRPORT <chr>
```

```
#Outra opção estranha

#intersect(airports$IATA_CODE, flights$DESTINATION_AIRPORT)
```

```
flights_left <- flights %>%
  left_join(airports %>%
              select(IATA_CODE, CITY, STATE),
```

```
            by = c("DESTINATION_AIRPORT" = "IATA_CODE")
            )
```

QUESTÃO 3

```
flights_left %>%
  group_by(STATE) %>%
  summarise(Quantidade = n()) %>%
  arrange(desc(Quantidade))
```

```
# A tibble: 54 x 2
   STATE Quantidade
   <chr>      <int>
 1 TX            24
 2 CA            22
 3 AK            19
 4 FL            17
 5 MI            15
 6 NY            14
 7 CO            10
 8 MN             8
 9 MT             8
10 NC             8
# i 44 more rows
```

```
#Outra opção
#flights %>%
#   count(STATE, sort = TRUE)
```

QUESTÃO 4

```
#?leaflet
#?addTiles
#?addMarkers
flights_map <- flights %>%
  left_join(airports %>%
              select(IATA_CODE, LATITUDE, LONGITUDE),
```

```
            by = c("DESTINATION_AIRPORT" = "IATA_CODE")
            )
```

```
this <- leaflet(flights_map) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE
  )
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-30 18:38:05 -03"
```